# TELCO CUSTOMER CHURN ANALYSIS

This project analyses a sample data set of 7,043 customers of a fictional telecom company that sells home phone and internet services. It looks at which customers leave (churn) and what is linked to it, using Excel pivot tables and a dashboard.

## BUSINESS QUESTION:

Which customer groups churn the most, and what can the company do to keep them?

## DATASET:

Telco Customer Churn, an IBM sample dataset on Kaggle (https://www.kaggle.com/datasets/blastchar/telco-customer-churn). It has 7,043 customers and 21 columns, and it is about a fictional company.

## TOOLS:

Excel: Pivot Tables, Formulas and charts.

## DATA CLEANING:

- Checked CustomerID for duplicates (No duplicates found)
- 11 rows had blank Total Charges. All had tenure 0 and had not churned (new customers, not yet billed), so I set the value to 0.
- Converted Senior Citizen from 0/1 to Yes/No.
- Added a Tenure Group column (0-12, 13-24, 25-48 and 49+ months) and a Churn Flag column (1 = churned, 0 = stayed).

## ANALYSIS:

Calculated churn rate (churned customers / total customers) by contract type, tenure group, internet service, senior citizen and payment method. Then checked whether the fiber, payment method and tenure patterns still held within each contract type, and compared average monthly charges for customers who left and stayed.

## KEY FINDINGS:

- 1,869 of 7,043 customers churned, a churn rate of 26.54%, so roughly 1 in 4 customers leave.
- Month-to-month customers churn at 42.71% (1,655 of 3,875), compared with 11.27% on one-year and 2.83% on two-year contracts. They make up about 89% of all churned customers.
- Customers in their first year churn at 47.44% (1,037 of 2,186), falling to 9.51% for those with 4+ years.
- Senior citizens churn at 41.68% (476 of 1,142), compared with 23.61% for non-seniors. They are about 16% of customers but about 25% of all churned customers.
- Contract type matters even among new customers. In the first year, 51.35% of month-to-month customers churn, compared with 10.48% on one-year contracts.

About 1,024 of the 1,869 churned customers (roughly 55%) were first-year month-to-month customers.

- Fiber optic customers churn more than DSL customers on every contract type (54.61% vs 32.22% on month-to-month, 19.29% vs 9.30% on one-year). So the fiber gap is not only because of contract.
- Electronic check users churn more than other payment methods on every contract type (53.73% vs 31.58-34.13% on month-to-month, 18.44% vs 6.82-10.30% on one-year). The gap is smaller than the overall comparison, so part of it comes from contract mix.
- Customers who churned paid somewhat higher monthly charges on average (74.44 vs 61.27). The difference is smaller than the contract effect, so price looks like a secondary factor.

## RECOMMENDATIONS:

- Offer discounts or benefits for moving month-to-month customers onto one- or two-year contracts.
- Give new customers extra attention in their first year, such as onboarding calls or early offers.
- Look into why fiber customers leave, for example price, speed or service quality.
- Encourage electronic check users to switch to automatic payment.
- Senior customers churn at nearly twice the rate of others, look into what they need (for example, simpler plans, easier support or better pricing) and target retention offers at them.
- Review pricing for high-paying customers with loyalty offers.
- Run exit surveys with customers who leave, to find out why.

## DASHBOARD:

